Econometrics

Estimation
Confidence Intervals
Hypothesis Testing

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Estimators (based on SW chapter 3)

- An *estimator* is a function of a sample of data. Given the sample is random, so is the estimator.
- An *estimate* is the numerical result **produced** by an estimator. It's a number, not a random variable.

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- An *estimator* is a function of a sample of data. Given the sample is random, so is the estimator.
- An *estimate* is the numerical result **produced** by an estimator. It's a number, not a random variable.
- Let us focus on estimating the population mean $E(Y) = \mu_Y$.
- Given a set of sample data, how could we estimate this quantity?

Estimators for $E(Y)$

- We already met $\bar{y}$ as a good estimator for $E(Y)$.
- But it's not the only option: We could also have used the average of the largest two extremes, like $\tilde{y} = \frac{max(y)-min(y)}{2}$, no? Or what about the *last* value in our i.i.d. sample of data $\{y_i\}_{i=1}^n$, y_n?
- generically, we write an estimator for population quantity x often $\hat{x}$.

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- generically, we write an estimator for population quantity x often $\hat{x}$.

What do we care about when choosing an estimator?

1. Unbiasedness
2. Consistency
3. Efficiency

Unbiasedness

- We have already heard that being unbiased means simply that

$$E(\hat{x}) = x,$$

for example we saw

$$E(\bar{y}) = \mu_Y.$$

- Is last value y_n an unbiased estimator for μ_Y?

Consistency

- We also heard already that we like estimators where some form of *law of large numbers* applies.

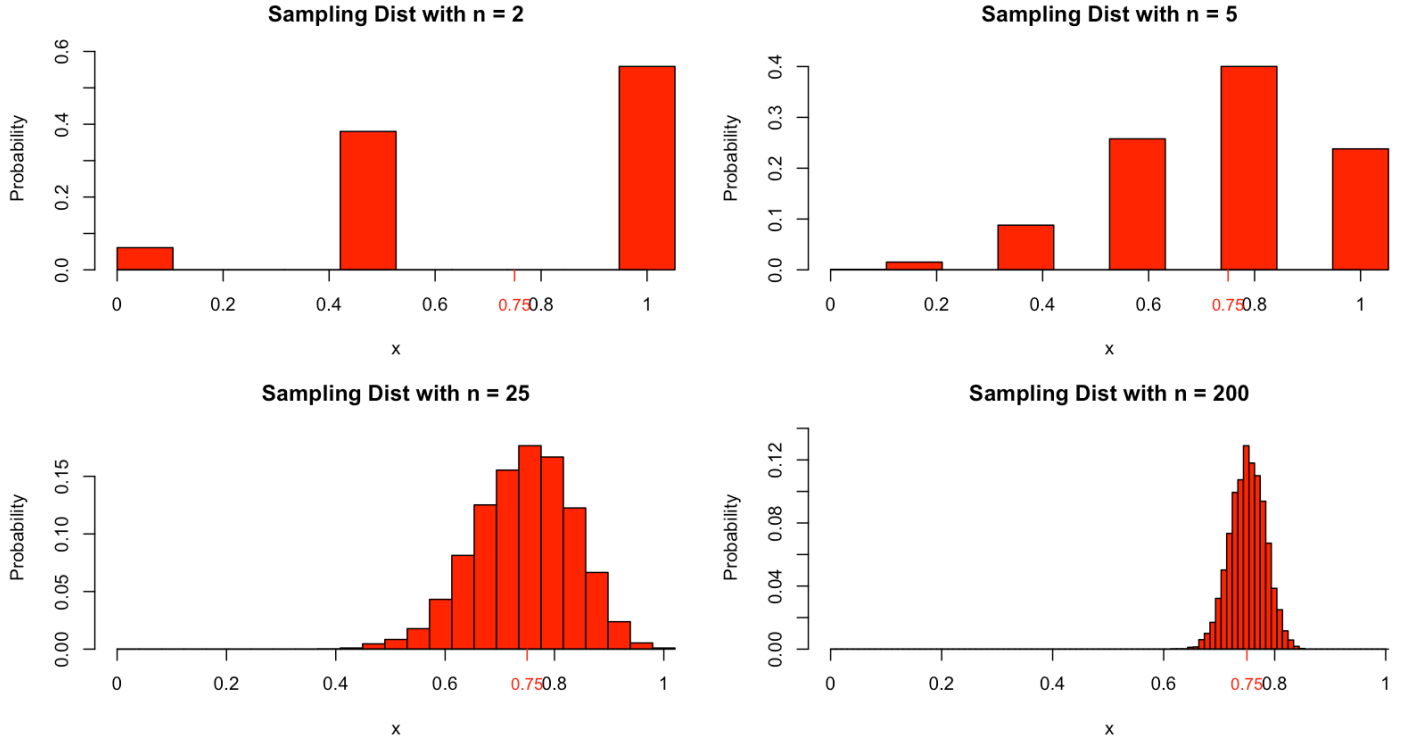
Consistency

- We also heard already that we like estimators where some form of *law of large numbers* applies.
- This is because we acknowledge that there is randomness in **any** estimate we could come up with.
- But ***at least*** we want to use the power of a lot of information (large n) to make sure that this randomness could be made arbitrarily small.

Consistency

- We also heard already that we like estimators where some form of *law of large numbers* applies.

- $\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$ depends on n
- We wrote $\bar{y} \xrightarrow{p} \mu_Y$
- Is this true? $y_n \xrightarrow{p} \mu_Y$? Why or Why not?



Efficiency

- If two estimators $\hat{x}_1$ and $\hat{x}_2$, both unbiased, have variances such that

$$var(\hat{x}_1) > var(\hat{x}_2)$$

then we say that $\hat{x}_2$ is *more efficient* than $\hat{x}_1$ - because it uses the same amount of data in a more efficient way.

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then we say that $\hat{x}_2$ is *more efficient* than $\hat{x}_1$ - because it uses the same amount of data in a more efficient way.

- Both $\bar{y}$ and y_n are unbiased. Which is more efficient?

Task: what about this estimator?

Define

$$\tilde{y} = \frac{1}{n}\left(\frac{1}{2}y_1 + \frac{3}{2}y_2 + + \frac{1}{2}y_3 + \cdots + \frac{3}{2}y_n\right)$$

and assume n is an even number.

1. Show that this is unbiased.
2. Show that this is consistent.
3. Show that the sample average is more efficient that this.

BLUE

- The estimator on the previous slide is very similar to the sample mean - just differently weighting the observations.
- We have seen that the main problem of tilde was that it has a large variance.
- So, among all unbiased estimators which are weighted averages of the Y's, the sample mean is the most efficient.
- We say ybar is the Best Linear Unbiased Estimator (BLUE).

Sample Variance and Standard Error

- We often don't know the variance of the population either - so we need to estimate it.
- We call s_Y^2 the *sample variance*, and it's an estimator for the population variance σ_Y^2

$$s_Y^2 = \frac{1}{n-1} \sum_{i=1}^{n} (y_i - \bar{y})^2$$

- this is quite similar to the population variance. What's different?

Degrees of Freedom

- We are dividing by $n - 1$ instead of n and we substract $\bar{y}$ instead of μ_Y

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- Well, we use $\bar{y}$ because we don't know μ_Y
- Let's show where the $(n-1)$ thing comes from

Task: Show where the $n - 1$ sample correction comes from

We want to show that this $\mathbb{E}\left[(Y_i - \overline{Y})^2\right]$ is not exactly the same as $\mathbb{E}\left[(Y_i - \mu_Y)^2\right]$, the latter being the population variance.

In particular, show that there is an $n - 1$ too much:

$$\mathbb{E}\left[\sum_{i=1}^{n}(Y_i - \overline{Y})^2\right] = n\sigma^2\left(1 - \frac{1}{n}\right) = (n-1)\sigma^2.$$

- You can see that for small n, 10 say, that makes quite a difference.

Standard error of $\bar{y}$

- The estimator for the standard *deviation* of $\bar{y}$ $\left(\frac{\sigma_y}{\sqrt{n}}\right)$ has a special name: we call it the **standard error**.
- We write $SE(\bar{y}) = \hat{\sigma}_{\bar{y}} = \frac{s_Y}{\sqrt{n}}$
- Again, observe that this is identical to $\left(\frac{\sigma_y}{\sqrt{(n)}}\right)$ (the std. deviation of $\bar{y}$) ***except*** for the fact that we cannot use σ_Y - because we often don't know it.
- s_Y is the estimator we derived on the previous slide!

Confidence Intervals

- A *Confidence Interval* (CI) for a given population parameter θ is a *random* interval whose endpoints are statistics (they are random variables)
- A CI contains θ with a preassigned probability.
- For example, a 95% confidence interval for θ is an interval in which θ lies 95% of the time.
- The CI is called an **interval estimate** (as opposed to a *point* estimate - which is what the sample mean is).

Confidence Intervals for the Population Mean

- Consider an i.i.d. sample Y_i with unknown mean μ_Y and **known** variance σ_Y^2.
- Standardizing any random variable X means

$$Z = \frac{X - E(X)}{sd(X)}, \qquad E(Z) = 0, \quad Var(Z) = 1$$

This guarantees mean 0 and variance 1, but does not imply that Z is normally distributed unless X itself is normal.

- For the sample mean, by the Central Limit Theorem (CLT), for large n we have

$$\frac{\bar{Y} - \mu_Y}{\sigma_Y / \sqrt{n}} \approx \mathcal{N}(0, 1)$$

and if the Y_i are normally distributed, this relationship holds **exactly** for any n.

Confidence Intervals for the Population Mean

- Thus, the standardized sample mean is approximately standard normal when σ_Y is known (exactly so if Y_i are normal).
- If σ_Y is **unknown**, we replace it with the sample standard deviation s_Y and obtain

$$T = \frac{\bar{Y} - \mu_Y}{s_Y/\sqrt{n}} \sim t_{n-1}$$

which converges to $\mathcal{N}(0,1)$ as $n \to \infty$.

Confidence Intervals for the population mean in large samples

- We want to find two numbers c_L, c_H to form an interval $[c_L, c_H]$ around μ with confidence $1-\alpha$:

$$\Pr\left(c_L \leq \mu \leq c_H\right) = 1-\alpha,$$

or

$$\Pr\left(\frac{\bar{y}-c_H}{s} \leq \frac{\bar{y}-\mu}{s} \leq \frac{\bar{y}-c_L}{s}\right) = 1-\alpha,$$

Confidence Intervals for the population mean in large samples

- We said $\frac{\bar{y}-E(Y)}{s} \approx \mathcal{N}(0,1)$ is approximately standard normal. *large sample* means we will assume it **is** normal.
- Also, we want this interval to be symmetric, so that

$$\bar{y} - c_H = -\left(\bar{y} - c_L\right)$$

- So, to compute this

$$\Pr\left(\frac{\bar{y} - c_H}{s} \leq \frac{\bar{y} - \mu}{s} \leq \frac{\bar{y} - c_L}{s}\right) = 1 - \alpha,$$

we do

$$\Phi\left(\frac{\bar{y} - c_L}{s}\right) - \Phi\left(-\frac{\bar{y} - c_L}{s}\right) = 1 - \alpha$$

Confidence Intervals for the population mean in large samples

Finally, using the symmetry of Φ,

$$\Phi\left(\frac{\bar{y}-c_L}{s}\right)-\Phi\left(-\frac{\bar{y}-c_L}{s}\right)=1-\alpha$$

becomes

$$2\Phi\left(\frac{\bar{y}-c_L}{s}\right)-1=1-\alpha$$

or

$$\Phi\left(\frac{\bar{y}-c_L}{s}\right)=1-\frac{\alpha}{2}$$

Confidence Intervals for the population mean in large samples

- we choose a value for α. Say $\alpha = 0.05$

$$\Phi\left(\frac{\bar{y} - c_L}{s}\right) = 1 - \frac{0.05}{2} = 0.975$$

- We need find the value for c_L! Easy - just invert the Φ function (or look up in your probability table for standard normal)

$$\frac{\bar{y} - c_L}{s} = \Phi^{-1}(0.975)$$

(`qnorm(0.975) = 1.959964` ≈ 1.96)

which we can easily solve for c_L now and get

$$c_L = \bar{y} - 1.96s, \quad c_H = \bar{y} + 1.96s$$

What if σ^2 is not known?!

- Then we just use a consistent estimator for it, i.e. we use $SE(\bar{y})$!
- Remember that this standard error depends on n, and so will our test statistic:

$$\frac{\bar{Y} - c_L}{S/\sqrt{n}} \sim t_{n-1}$$

- for small n this is now t distributed - have to work with t-dist table instead of Φ as above.
- For any $n > 30$, can use the normal again.

Demo

```r
library(shinyCLT)
CLT()
```

Packages used in this set of slides

```
library(tidyverse)
library(infer)
library(moderndive)
```

Is There Gender Discrimination In Promotions?

- Article published in the *Journal of Applied Psychology* in 1970 investigates whether female employees at Banks are discriminated against.
- 48 supervisors were given *identical* candidate CVs - identical up to the first name, which was male or female.
- Many similar experiments have been conducted with other groups. Arabic Names, Black names, Jewish names or other groups that can be identified from typical name choice.

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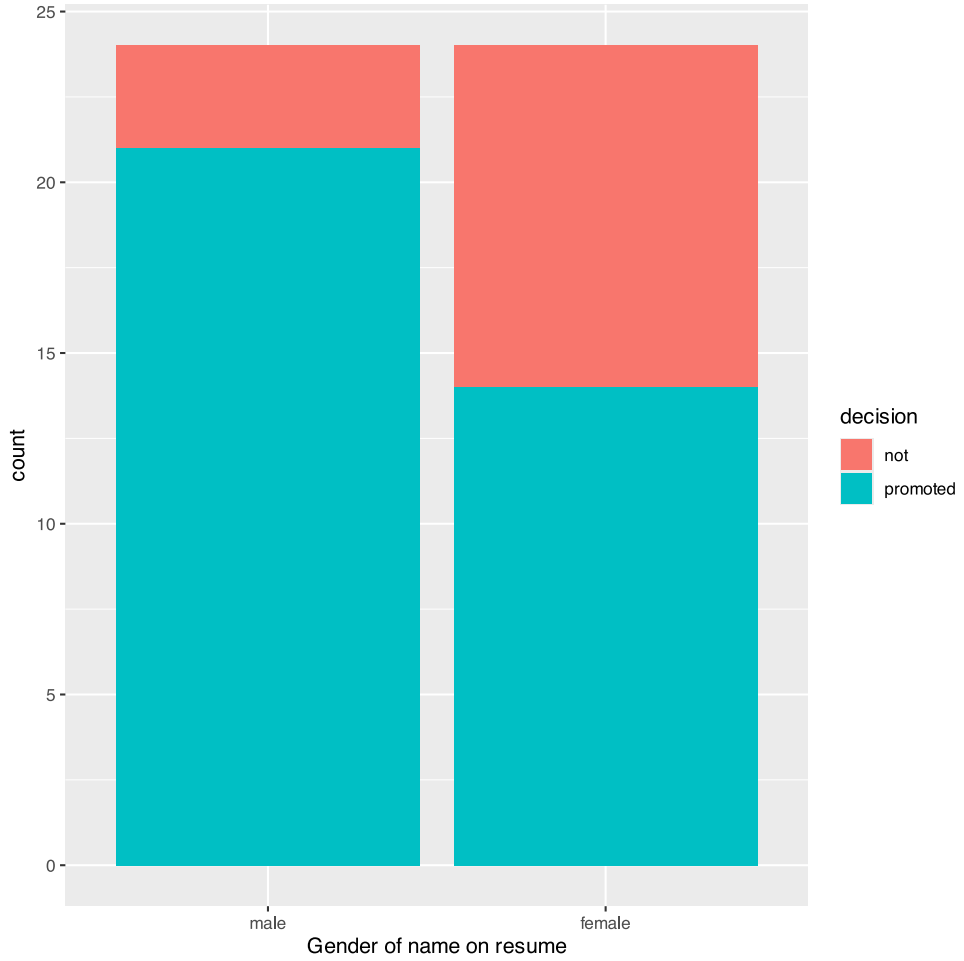
```r
library(moderndive)
promotions
```

```
## # A tibble: 48 × 3
##       id decision gender
##    <int> <fct>    <fct>
##  1     1 promoted male
##  2     2 promoted male
##  3     3 promoted male
##  4     4 promoted male
##  5     5 promoted male
##  6     6 promoted male
##  7     7 promoted male
##  8     8 promoted male
##  9     9 promoted male
## 10    10 promoted male
## # ℹ 38 more rows
```

Looking At Promotions




```
promotions %>%
  group_by(gender, decision) %>%
  summarize(n = n()) %>%
  mutate(proportion = n / sum(n))
```

```
## # A tibble: 4 × 4
## # Groups:   gender [2]
##   gender decision     n proportion
##   <fct>  <fct>    <int>      <dbl>
## 1 male   not          3      0.125
## 2 male   promoted    21      0.875
## 3 female not         10      0.417
## 4 female promoted    14      0.583
```

- 87.5% of "men" were promoted.
- 58.3% of "women" were promoted.
- That's a difference of 87.55 - 58.3% = 29.2%.
- Is the 29% advantage for men in this sample **conclusive evidence**?
- In a *hyopthetical world* **without gender discrimination**, could we have observed a 29% difference *by chance*?

Imposing A Hypothetical World: No Gender Discriminiation

- Suppose we lived in a world without gender discrimination.
- The label `gender` in our dataframe would be meaningless.
- Let's randomly reassign `gender` to each row and see how this affects the result.
- Suppose we have 48 playing cards: 24 red (female) and 24 (black)
- Shuffle the cards, and lay down the cards in a row, record `f` if **red**.

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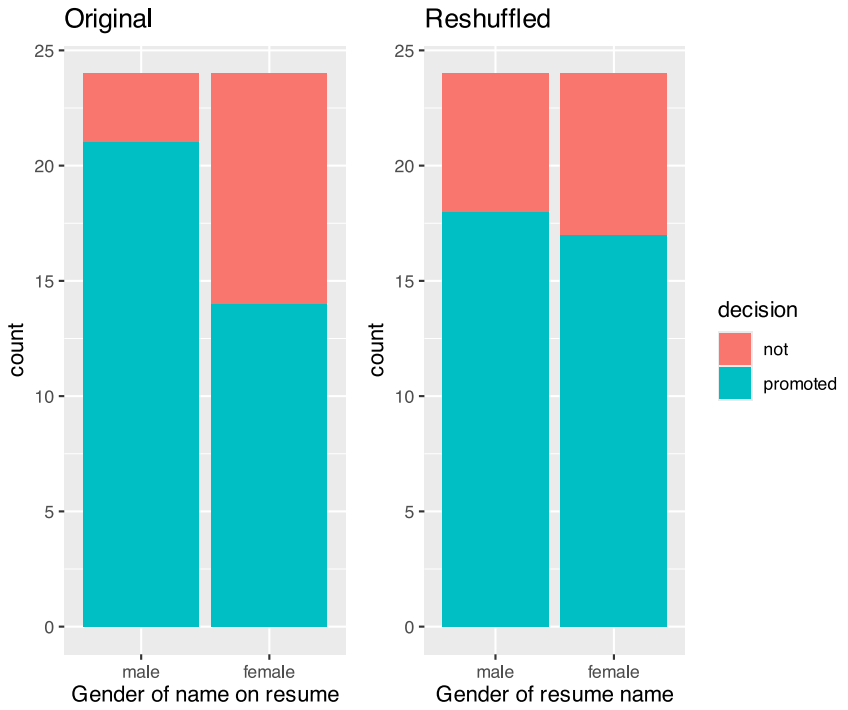
- Shuffle the cards, and lay down the cards in a row, record `f` if **red**.

```
bind_cols(promotions, dplyr::select(promotions_shuffle
```

id	decision	gender	decision_shuffled	gender_s
1	promoted	male	promoted	female
2	promoted	male	promoted	female
3	promoted	male	promoted	male
4	promoted	male	promoted	female
5	promoted	male	promoted	male
6	promoted	male	promoted	male
7	promoted	male	promoted	male
8	promoted	male	promoted	female
9	promoted	male	promoted	male
10	promoted	male	promoted	female
11	promoted	male	promoted	male

Reshuffled Promotions




```
promotions %>%
  group_by(gender, decision) %>%
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  mutate(proportion = n / sum(n))
```

```
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## # Groups:   gender [2]
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## 3 female not         10      0.417
## 4 female promoted    14      0.583
```

```
promotions_shuffled %>%
  group_by(gender, decision) %>%
  summarize(n = n()) %>%
  mutate(proportion = n / sum(n))
```

```
## # A tibble: 4 × 4
## # Groups:   gender [2]
##   gender decision     n proportion
##   <fct>  <fct>    <int>      <dbl>
## 1 male   not          6      0.25
## 2 male   promoted    18      0.75
## 3 female not          7      0.292
## 4 female promoted    17      0.708
```

Sampling Variation?

- In the hypothetical world, the difference was only 4.2%.
- But what's the role of *sampling variation*? How representative of that hypothetical world is 4.2%?
- Let's construct the sampling distribution ourselves!

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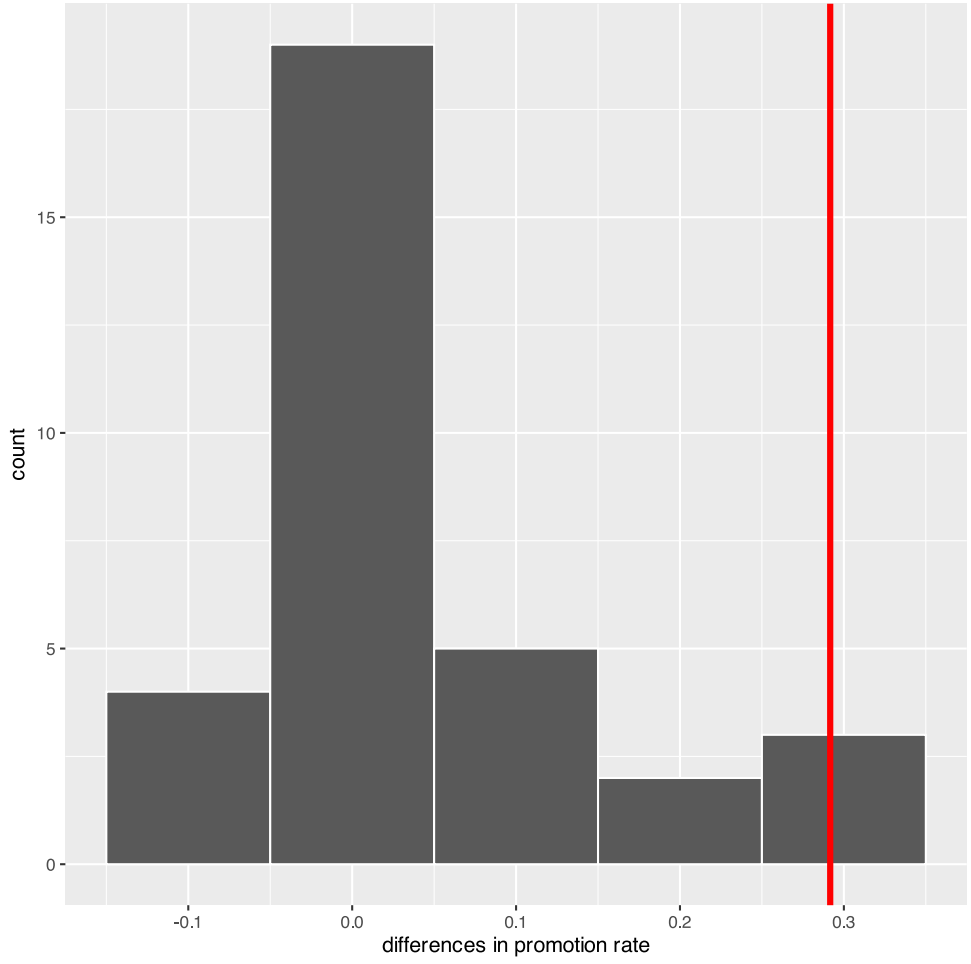
- Let's construct the sampling distribution ourselves!

1. You need to shuffle a deck of 48 cards, 24 red, 24 black, and lay out card after card in front of you.
2. You do **not** put the cards back into the deck!
3. You could use the function `sample` for example. Look at `?sample` to find out more.
4. fill in your results into this shared spreadsheet!

Sampling Variation in Reshuffling



- This distribution was created in our **hypothetical** scenario: no discrimination.
- We see how sampling variation affects the difference in promotion rates.
- The red line denotes the *observed difference* in the **real world**.
- Now: How *likely* is it that the red line is part of this **hypothetical** distribution?

Recap

- We just did a **permutation test**. We randomly reshuffled and checked if it makes a difference.
- Again Resampling: Boostrapping is with replacment, permutation is without.
- Bootstrapping: we put the paper slips **back** after recording them.
- Permutation: We took card after card from our deck (*without* putting it back!)

Recap

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- Bootstrapping: we put the paper slips **back** after recording them.
- Permutation: We took card after card from our deck (*without* putting it back!)

- We observed the estimate $\hat{p}_m - \hat{p}_f = 29\%$ in the real world.
- We *tested* whether in a hypothetical universe with no discrimination, 29% *likely* to occur.
- We concluded *rather not*. We tended to **reject** that hypothesis.
- The real question was: is 29% **really** different from zero? What is the role of sampling variation?

Hypothesis Testing Setup

Hypothesis Test Notation and Definitions

- In Hypothesis testing we compare two **competing hypothesis**.
  - In our example:

$$H_0 : p_m - p_f = 0$$
$$H_A : p_m - p_f > 0$$

  - H_0 stands for the **null hypothesis**, where *no effect* is observed. That's our hypothetical world from above.
- H_A or H_1 is the **alternative** hypothesis. Here, we have a *one-sided* alternative, saying that $p_m > p_f$, ie women are discriminated against. The *two-sided* formulation is just $H_A : p_m - p_f \neq 0$

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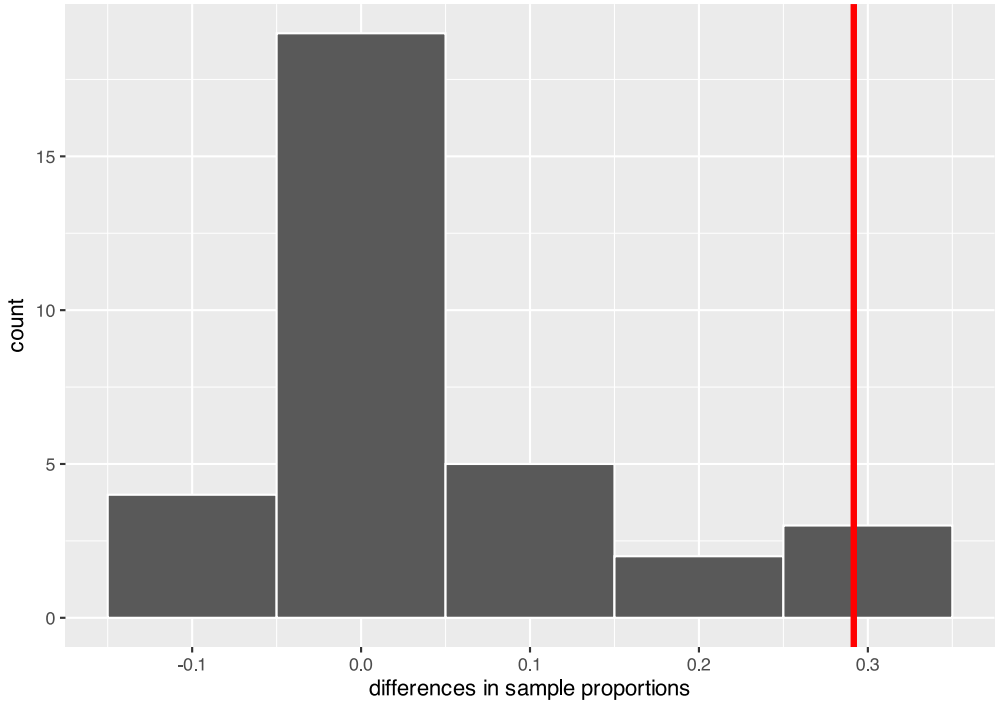
- A **test statistic** is a summary statistic which we use to summarise a certain aspect of our sample. Here: $\hat{p}_m - \hat{p}_f$
- The *observed test statistic* is the number we get from our real world sample: $\hat{p}_m - \hat{p}_f = 29\%$
- The **null distribution** is the sampling distribution of our test statistic, assuming the Null hypothesis is **true**. That's our hypothetical world without discrimination.
- We have seen such a null distribution just above:

Null Distribution



- This **is** the sampling distribution of $\hat{p}_m - \hat{p}_f$, assuming H_0 is true.
- The red line is the *observed* test statistic.

P-Value and Significance Level α

- The **p-value** is the probability of observing a test statistic *more extreme* than the one we obtained, assuming H_0 is true. 🤔
- How *strong* a piece of evidence is it to observe $\hat{p}_m - \hat{p}_f = 29\%$ in a world where $p_m - p_f = 0$ is assumed true? Very strong? Not so strong?
- How many samples did we obtain that had a difference *greater* than 29%? Many, or not so many?
- The p-value quantifies this by measuring the probability to the right of the red line in the previous plot.

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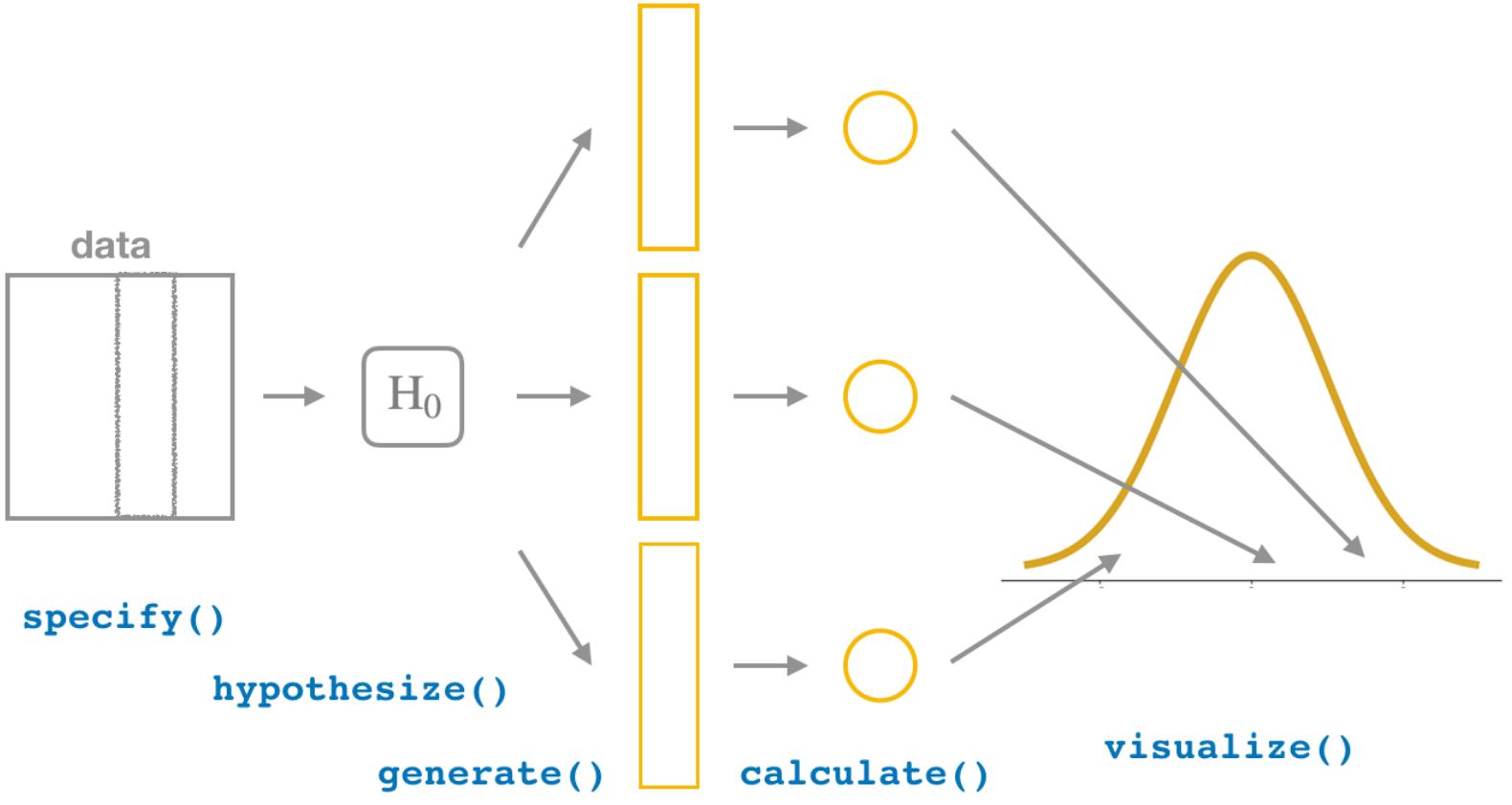
- How many samples did we obtain that had a difference *greater* than 29%? Many, or not so many?
- The p-value quantifies this by measuring the probability to the right of the red line in the previous plot.

- The **significance level** α is a *cutoff* on the p-value.
- We choose it *before* conducting our hypothesis test. It's common to assume $\alpha = 5\%$.
- If the p-value falls below the cutoff α, we **reject** the null hypothesis on the grounds that *what we observe is too unlikely to happen* under the Null.
- Small p-value: The red line is *too far* from the center of the Null distribution. Observing the red line would have happened with very small probability only.

Conducting Hypothesis Tests

Testing with `infer`



`infer` Testing Pipeline

- Here we follow closely the infer workflow given in moderndive.
- We augment our previous pipeline with the `hypothesize` function, defining the type of null hypothesis.
- Also, we give a `formula` to `specify()` this time, instead of only a variable name as before.
- We create the Null Distribution by *reshuffling* (deck of cards), and *not* by *resampling* (pennies).

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- We augment our previous pipeline with the `hypothesize` function, defining the type of null hypothesis.
- Also, we give a `formula` to `specify()` this time, instead of only a variable name as before.
- We create the Null Distribution by *reshuffling* (deck of cards), and *not* by *resampling* (pennies).

```r
null_distribution <- promotions %>%
  # takes formula, defines success
  specify(formula = decision ~ gender,
          success = "promoted") %>%
  # decisions are independent of gender
  hypothesize(null = "independence") %>%
  # generate 1000 reshufflings of data
  generate(reps = 1000, type = "permute") %>%
  # compute p_m - p_f from each reshuffle
  calculate(stat = "diff in props",
            order = c("male", "female"))
null_distribution
```

```
## Response: decision (factor)
## Explanatory: gender (factor)
## Null Hypothesis: ind...
## # A tibble: 1,000 × 2
##    replicate    stat
##        <int>   <dbl>
##  1         1 -0.0417
##  2         2  0.125
##  3         3 -0.0417
##  4         4 -0.125
##  5         5  0.208
##  6         6 -0.125
##  7         7 -0.125
##  8         8 -0.208
##  9         9  0.208
## 10        10 -0.125
```

Back to Reality: What did we *Observe*?

- We computed $\hat{p}_m - \hat{p}_f$ from our *real-world* sample before.

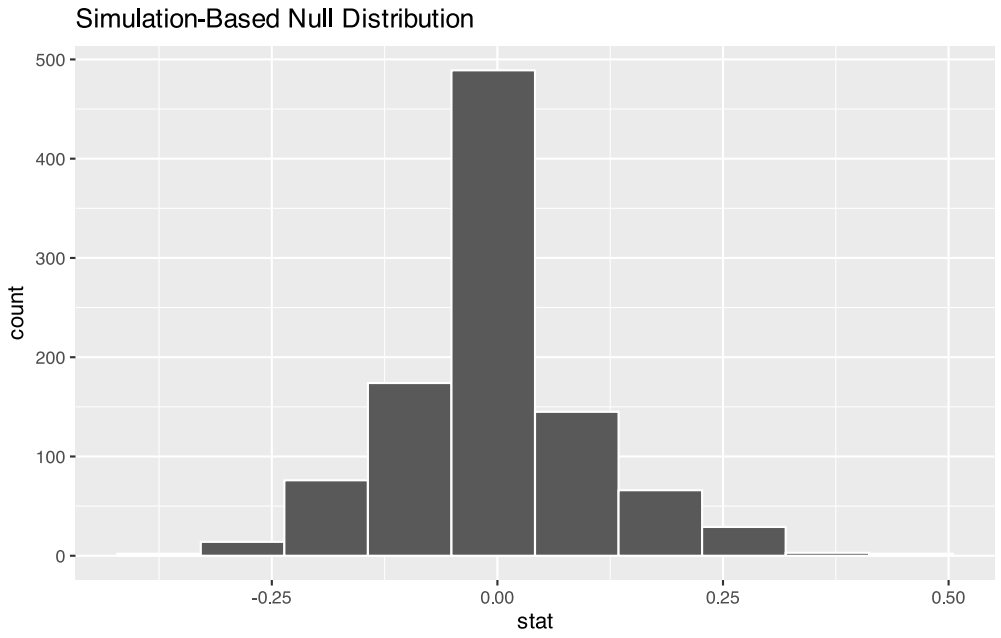
```
obs_diff_prop <- promotions %>%
  specify(decision ~ gender, success = "promoted") %>%
  calculate(stat = "diff in props", order = c("male",
obs_diff_prop
```

```
## Response: decision (factor)
## Explanatory: gender (factor)
## # A tibble: 1 × 1
##    stat
##   <dbl>
## 1 0.292
```

- How does that observed statistic compare the distribution of **this** test statistic, assuming that H_0 is true?
- We **created** that distribution on the previous slide: `null_distribution`.
- Let's confront `null_distribution` with `obs_diff_prop`, and let's compute the p-value!

Visualize the Null

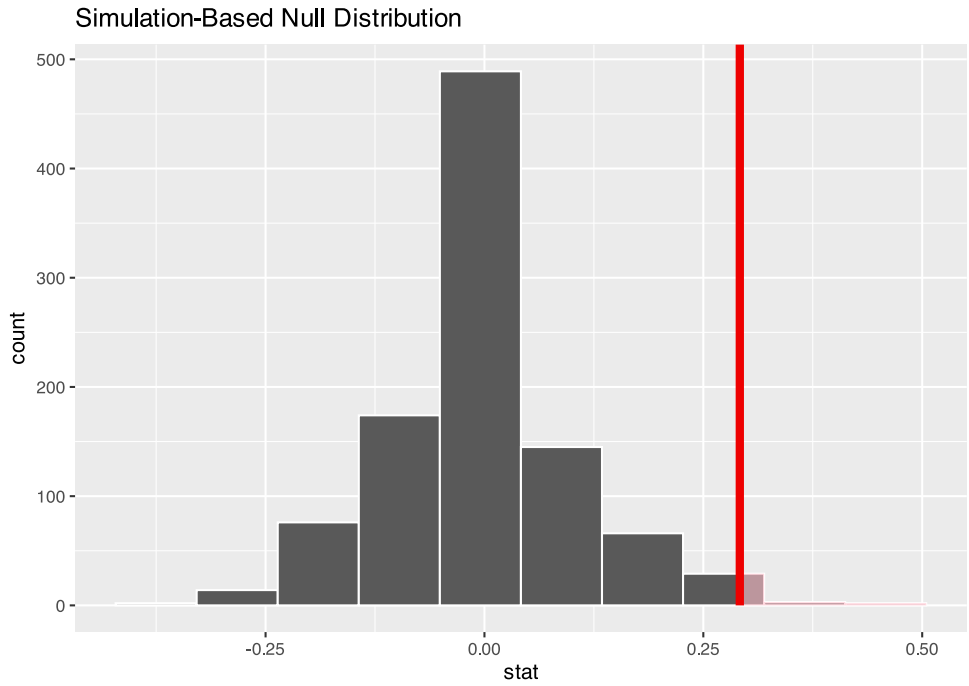
```
visualize(null_distribution, bins = 10)
```



- This is the distribution of $\hat{p}_m - \hat{p}_f$ under H_0 .
- No Discrimination in that world.

Visualize the P-value

```
visualize(null_distribution, bins = 10) +
  shade_p_value(obs_stat = obs_diff_prop,
                direction = "right")
```



- `shade_p_value` adds the p-value based on `obs_diff_prop`, i.e 0.29.
- `direction = "right"` represents our one-sided alternative $H_A : p_m - p_f > 0$
- *more extreme* means *bigger difference* here, hence *more to the right*.
- If $H_A : p_m - p_f < 0$, we'd set `direction = "left"`
- The red area **is the p-value**!
- Is that a *big* or a *small* area?

Obtaining the p-value and Deciding to Reject

- Obtain the precise p-value with

```
p_value <- null_distribution %>%
  get_p_value(obs_stat = obs_diff_prop, direction
p_value
```

```
## # A tibble: 1 × 1
##   p_value
##     <dbl>
## 1   0.034
```

- So, the probability of observing a 29% difference in a world with no discrimination is only 3.4%. That probability is due to sampling variation.

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- So, the probability of observing a 29% difference in a world with no discrimination is only 3.4%. That probability is due to sampling variation.

- Suppose we had set $\alpha = 0.001 = 0.1\%$
- Given that the p-value is *greater* than α,
  - i.e. 3.4% > 0.1%,
  - we would **fail to reject** the null $H_0 : p_m - p_f = 0$.
- The p-value was not sufficiently small to convince us in this case.
- What would have happened, had we set cutoff $\alpha = 0.05 = 5\%$ instead?

Testing Errors

- Working with probabilities implies that sometimes, we make an error.
- 29% may be *unlikely* under H_0, but that doesn't mean it's *impossible* to occur.
- So, it may happen that we sometimes reject H_0, when in fact it was true.

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- This is similar to a verdict reach in a court trial:

	Truly not guilty	Truly guilty
Verdict		
Not guilty verdict	Correct	Type II error
Guilty verdict	Type I error	Correct

- In fact, in hypothesis testing:

	H0 true	HA true
Verdict		
Fail to reject H0	Correct	Type II error
Reject H0	Type I error	Correct

Type I and Type II Errors

- So, there are even two types of errors to make! 😯
- Type I: We convict an innocent person. We Reject a *true* Null.
- Type II: We *fail* to convict a criminal. We *fail* to reject a *wrong* Null.
- We **choose** the frequency of a Type I error by setting α, called the **significance level**.

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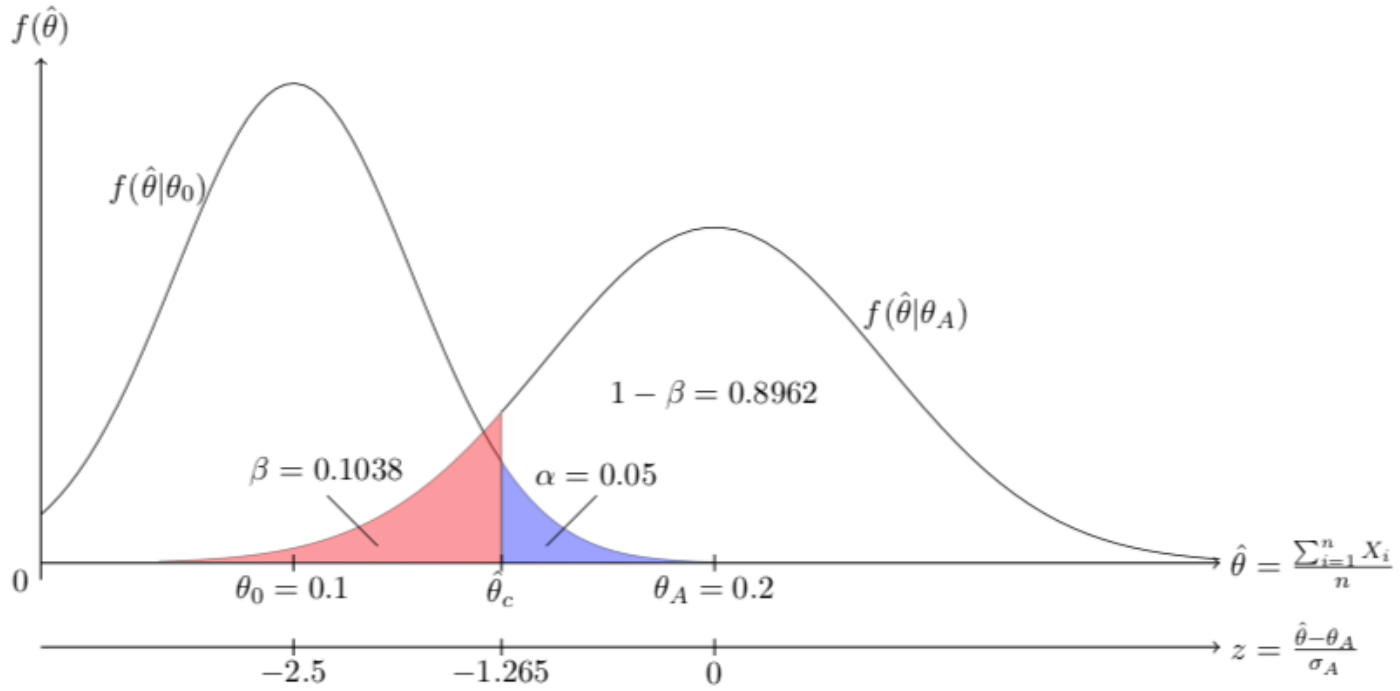
- We **choose** the frequency of a Type I error by setting α, called the **significance level**.

- The probability of committing a type II error is called β. The value $1 - \beta$, i.e. the prob. of *not* making such an error, is called the **power** of a hypothesis test.
- Ideally, $\alpha = \beta = 0$. However, with random sampling this is impossible. Also, both errors are inversely related. (see next slide)
- So, typically we fix α and try to maximize the power of the test.
- Given a certain frequency of convicting an innocent person, we try to make sure we convict as many true criminals as possible.

Type I and II Errors are Inversely related



- $\hat{\theta}$ is *some* test statistic.
- $f(\hat{\theta}|\theta_0)$ and $f(\hat{\theta}|\theta_A)$ are Null and Alternative distributions.
- Changing α moves critical value $\hat{\theta}_c$.
- This example is fully worked out here

THANKS

To the amazing moderndive team!

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END

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